

Sign-Aware Antithetic Coupling for Signed Gradient-Particle Methods

Stephen Abkin · Prabir Daripa

Abstract Gradient random-walk methods represent the spatial derivative of a scalar solution by signed particles and recover the field by cumulative summation. When such a field is estimated by averaging repeated simulations, the accuracy of the average depends on how the random increments of the simulations are coupled. At a fixed diffusion stage, reflecting the Gaussian increments maximizes a matched pair's conditional variance contribution when its masses have opposite signs. We therefore reflect matched increments when the masses agree in sign and synchronize them when they do not. For a single matched pair this rule minimizes the covariance of the two contributions pointwise, and it leaves the discrete law of each simulation exactly unchanged, so the average retains the bias of the underlying solver. We derive an exact expression for the resulting reduction of conditional integrated variance at one diffusion stage, and we identify the configurations in which the rule provably does nothing. Numerical experiments on viscous Burgers and cubic-flux problems compare the coupling with reflected pairing, with pairing restricted to each sign class, with conditional expectation and control variates, and with a randomized quasi-Monte Carlo adaptation. Against within-sign pairing the integrated field variance falls to 0.739 [0.682, 0.801] of its value at comparable cost, and in an experiment with plans selected before evaluation the coupled estimator reaches a prescribed expected-error target using four solver trajectories against six.

Keywords gradient random walk · signed particles · antithetic coupling · variance reduction · viscous conservation laws · Monte Carlo

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S. Abkin

Department of Mathematics, Texas A&M University, College Station, TX, USA
E-mail: stephen122204@tamu.edu

P. Daripa

Department of Mathematics, Texas A&M University, College Station, TX, USA
E-mail: daripa@tamu.edu

1 Introduction

Monte Carlo methods offer a way to approximate solutions of partial differential equations through averages of simulated trajectories. With independent samples and finite variance, the standard error decays as the inverse square root of the sample count. The variance and the cost of a sample still depend on the problem, so reducing either can make a substantial difference to the work needed for a given accuracy. Among the Monte Carlo methods developed for one-dimensional parabolic and viscous problems, one family replaces the solution by its spatial derivative and carries that derivative on signed particles [10, 21–24]. The field is then recovered as a cumulative sum of particle masses rather than as a histogram. Because particles accumulate where the derivative is large, the representation resolves fronts and steep layers automatically, and the reconstruction requires no grid. These gradient random-walk methods have been analyzed for viscous scalar conservation laws [3, 5–7, 14, 21, 25], and have recently been extended by Bertaglia, Pareschi and Caflisch [4] to relaxation approximations of hyperbolic conservation laws, including systems.

We are concerned with a question that arises once such a solver has been chosen. A single simulation returns a field with sampling error; the usual remedy is to average several independent simulations, at a cost proportional to their number. Antithetic sampling seeks to improve the accuracy obtained from the same number of simulations: run two simulations whose random inputs are negatively related, so that their errors partly cancel in the average, while each simulation individually retains its correct law [11, 12]. For a particle solver the natural implementation is immediate. The solver already sorts its particles by position at every step, so the two simulations can be matched at equal spatial rank, and the matched Gaussian diffusion increments can be reflected, $Z \mapsto -Z$.

The signed representation gives us a reason to reconsider that choice at each matched pair. Reflecting the increments of two matched particles makes their positions move in opposite directions. In a cumulative reconstruction, a particle contributes its mass to the field to the right of its position, so moving two particles apart moves their contributions apart only if the two masses have the same sign. If the matched masses have opposite signs, the same reflection makes their contributions to the field move *together*, and the covariance that antithetic sampling is meant to make negative becomes positive instead. Section 3.1 makes this precise: at a fixed pre-stage state, reflection minimizes the covariance of the two contributions when the masses agree in sign and maximizes it when they do not, and in each case the extremum is attained over all couplings of the two increments of one matched pair. This suggests a simple modification. We keep the matching by spatial rank and let the mass signs choose the orientation of the increment, reflecting when the signs agree and synchronizing when they differ.

There is another way to use the mass signs: keep reflecting every matched pair, but restrict the matching so that particles are only ever paired with particles of the same sign. This within-sign rule never creates an unfavorable

pair. It also gives up the spatial matching, since the r th positive particle of one simulation need not lie near the r th positive particle of the other. Which construction is better is not decidable by inspection, and the comparison between the two is the central empirical question of this paper. Reflected pairing enters as the common baseline against which both are measured.

The construction builds on established coupling ideas. Nüsken and Pavliotis [19] develop a general framework for coupling copies of a sampler while preserving marginals, and show that the efficient coupling switches between reflected and synchronous noise according to the sign of a product; their Proposition 56, Eq. (141), makes the switch depend on $\varphi'(x)\varphi'(y)$, where φ solves a Poisson equation associated with the observable, and their Corollary 60, Eq. (144), gives a Poisson-free special case in which the switch depends on $x \cdot y$ alone. Their Section 5.2, Eqs. (159)–(160), sorts the copies at every step, by $|\nabla\varphi|$, and couples adjacent members of the sorted list. Their analysis concerns the asymptotic variance of an ergodic, non-interacting sampler, and does not supply a finite-time comparison for the interacting solver considered here, but the switching principle and the repeated sorted matching are theirs. What is specific to the present setting is the choice of keys: the sort key is the particle position, which the solver computes anyway for the reconstruction, and the switch key is the gradient mass the particle already carries.

Within the gradient-particle literature, Lécot [15] combines signed cumulative reconstruction, re-sorting of the particles at every step, and quasi-random inputs to the diffusion stage for a Burgers problem; a related treatment appears in Lécot and Ogawa [16]. This is the closest precedent for structuring the random inputs of this class of solver, and our quasi-Monte Carlo comparison adapts his rank-and-increment map rather than reproducing his complete algorithm. Antithetic ideas have also been applied to mean-field particle systems by Bencheikh and Jourdain [2], who couple a $2N$ -particle system with two N -particle systems in order to cancel a particle-number bias. That construction compares systems of different sizes, whereas we average two systems of the same size, each carrying the same bias. Recent quasi-Monte Carlo constructions for McKean–Vlasov dynamics [1] and array-RQMC for a Fokker–Planck particle model [18] couple randomness within an interacting ensemble under assumptions different from ours.

The paper is organized as follows. Section 2 recalls the signed gradient-particle discretization, fixes the update whose law we shall preserve, and defines the estimators and the error functional. Section 3 develops the coupling: the pairwise calculation that dictates the switching rule, the admissible class in which all three pairings sit, the preservation of each simulation’s law and its consequence for the bias, an exact expression for the variance reduction obtained at one diffusion stage, and the configurations in which the construction provably does nothing. Section 4 reports the numerical comparisons: against the two alternative pairings, against conditional expectation and control variates applied to the same solver, across initial profiles, observation times and fluxes, at prescribed accuracy targets with plans frozen before evaluation, and for nonlinear functionals and the regime in which discretization bias dominates.

Section 5 concludes. The proofs, the test problems and reference solutions, the supporting estimators, the statistical methodology and the evidence provenance are collected in the appendices.

Throughout, the computations are one-dimensional and scalar, the particle masses are fixed at initialization, and transport uses the reconstructed velocity rather than a sampled velocity label. We compare estimators within this solver and do not compare the solver with deterministic alternatives.

2 Signed Gradient Particles for Viscous Conservation Laws

This section fixes the discretization whose law the coupling of Section 3 will preserve. Nothing in it is new; we state it in full because the coupling rule, the law-preservation argument, and the variance identity all refer to specific details of the update.

2.1 From the Field to its Gradient

We consider the scalar viscous conservation law

$$\partial_t u + \partial_x f(u) = \nu \partial_{xx} u, \quad u(x, 0) = u_0(x), \quad x \in \mathbb{R}, \quad 0 < t \leq T, \quad (2.1)$$

with $f \in C^1(\mathbb{R})$ and $\nu > 0$. The initial datum u_0 has bounded variation and integrable derivative, and the limits $u_{\pm} = \lim_{x \rightarrow \pm\infty} u_0(x)$ exist. Our experiments use $f(u) = u^2/2$ (viscous Burgers), $f(u) = u^3/3$, and $f \equiv 0$; the last is the heat equation and serves as a control in which the coupling acts in the absence of transport.

Let $w = \partial_x u$. Differentiating (2.1) in x for a smooth solution and using $\partial_x f(u) = f'(u) \partial_x u = f'(u)w$ gives

$$\partial_t w + \partial_x (f'(u) w) = \nu \partial_{xx} w, \quad w(x, 0) = u'_0(x). \quad (2.2)$$

Equation (2.2) is a conservation law for w with velocity field $f'(u)$ and diffusivity ν : the gradient is advected at the local characteristic speed and spread by diffusion. It is therefore natural to approximate w by point masses that are transported and diffused, which is the idea underlying gradient random-walk methods [10, 21, 22].

Accordingly we represent w by N signed particles with positions X_i and fixed masses m_i , and recover u by integrating w from the left,

$$w^N(\cdot, t) = \sum_{i=1}^N m_i \delta_{X_i(t)}, \quad U^N(x, t) = u_- + \sum_{i: X_i(t) \leq x} m_i, \quad (2.3)$$

where δ_{X_i} is the Dirac measure at X_i . The second expression in (2.3) is the cumulative sum that gives the method its character: every particle to the left of x contributes its mass to the field at x , so the reconstruction is a signed

step function whose jumps sit at the particle positions. It is the discrete analog of $u = u_- + \int_{-\infty}^x w$, and the same construction, with ± 1 masses and a left reconstruction, appears in Lécot [15, Eqs. (35)–(37)].

Two consequences of (2.3) are used repeatedly. First, letting $x \rightarrow +\infty$ gives

$$\sum_{i=1}^N m_i = u_+ - u_-, \quad (2.4)$$

so the total signed mass is fixed by the far-field data, exactly as in Lécot [15, Eq. (36)]. For every profile with equal far-field values this total vanishes. Second, the response of $U^N(x, \cdot)$ to a displacement of particle i has the sign of m_i : moving a positive-mass particle to the right removes mass from the field to its left, while moving a negative-mass particle to the right adds field there. This elementary observation is what makes the sign of a mass, and not only the position of a particle, relevant to the design of a coupling.

We initialize deterministically, so that both simulations of a pair start from an identical state. For the main experiments we place $N/2$ particles at midpoint quantiles of the positive part of u'_0 and $N/2$ at those of the negative part and assign magnitude

$$|m_i| = \frac{\text{TV}(u_0)}{N}, \quad \text{TV}(u_0) = \int_{\mathbb{R}} |u'_0(x)| \, dx. \quad (2.5)$$

Equal counts and equal magnitudes then give $\sum_i m_i = 0$ exactly. The profile/time study of Section 4.4 instead normalizes each sign class by its own sampled integral, and the one-sign control of Section 4.7 has all masses of one sign; both constructions are specified in Appendix C. We flag in Section 3.4 the one place where a vanishing total mass is used.

2.2 One Step of a Simulation

Fix a time step $h > 0$, let $K = T/h \in \mathbb{N}$, and set

$$\sigma = \sqrt{2\nu h}, \quad (2.6)$$

the standard deviation of the Brownian displacement accumulated over one step by the diffusion in (2.2) [10]. A simulation carries the state (X, m, v) of positions, masses, and the velocities to be used by the next transport. One step is the operator splitting of (2.2) into transport and diffusion, given as Algorithm 1.

The velocity step deserves comment, since it is where (2.3) re-enters. After the sort, the cumulative sum $u_- + \sum_{j \leq i} m_j$ is exactly U^N evaluated at the position of particle i , inclusive of that particle's own mass; the velocity assigned to particle i is therefore f' of the reconstructed field at that particle. Because the reconstruction is a step function, an inclusive and an exclusive convention differ by one mass, and we record that we use the inclusive one at every rank

Algorithm 1: One step of a single simulation.

Input: state (X, m, v) , step h , viscosity ν , flux derivative f'

- 1 transport: $X_i \leftarrow X_i + v_i h$ for $i = 1, \dots, N$;
- 2 sort: $\pi \leftarrow$ a stable sort of X , $(X, m) \leftarrow (X_\pi, m_\pi)$;
- 3 velocity: $v_i \leftarrow f'(u_- + \sum_{j \leq i} m_j)$;
- 4 diffusion: draw $Z \sim \mathcal{N}(0, I_N)$ independent of the past, $X_i \leftarrow X_i + \sigma Z_i$;

and in every coupling. This is a mean-transport update: the velocity is read off the reconstruction rather than sampled from a two-speed equilibrium, and we make no claim here about the relative merits of the two choices.

Three further details are used later. The velocity computed at step 3 is not consumed until the transport of the following step, so it is part of the state and must travel with its particle under any reordering. The sort is stable, so coincident particles keep their previous relative order. Finally, the positions are not re-sorted between the diffusion of one step and the transport of the next; the sort at step 2 does that work.

2.3 Estimators and Error

Write $U^{(A)}, U^{(B)}$ for the fields reconstructed from two simulations at time T . Averaging B independent simulations gives the unpaired estimator

$$\hat{u}_B^{\text{sing}}(x) = \frac{1}{B} \sum_{b=1}^B U^{(b)}(x). \quad (2.7)$$

When the simulations are run in coupled pairs we average within a pair first,

$$\bar{U}^{(b)}(x) = \frac{1}{2} (U^{(A,b)}(x) + U^{(B,b)}(x)), \quad (2.8)$$

and then over B independent realizations of the pair,

$$\hat{u}_B(x) = \frac{1}{B} \sum_{b=1}^B \bar{U}^{(b)}(x). \quad (2.9)$$

Thus B counts simulations in (2.7) and pairs in (2.9), and a paired estimate at a given B costs $2B$ solver trajectories. We report trajectory counts explicitly wherever the two conventions meet.

Accuracy is measured by the expected integrated squared error on a window $\Omega = [-5, 5]$, discretized at $L = 400$ equispaced points x_ℓ with spacing Δx ,

$$\mathcal{E}(B) = \mathbb{E} \|\hat{u}_B - u_{\text{ref}}\|_\Omega^2, \quad \|g\|_\Omega^2 = \Delta x \sum_{\ell=1}^L g(x_\ell)^2, \quad (2.10)$$

with u_{ref} an independently computed spectral or Cole–Hopf solution (Appendix C). Since the B realizations are independent and identically distributed, (2.10) splits as

$$\mathcal{E}(B) = b^2 + \frac{V}{B}, \quad b^2 = \|\mathbb{E}\bar{U} - u_{\text{ref}}\|_{\Omega}^2, \quad V = \Delta x \sum_{\ell=1}^L \text{Var}(\bar{U}(x_{\ell})). \quad (2.11)$$

The squared bias b^2 collects the finite-particle and time-stepping errors of the underlying discretization, together with any error in the reference; the integrated variance V is the only term that responds to averaging more realizations at fixed N and h . A coupling can act on V alone. This is not a limitation peculiar to the present construction, but it does fix what any comparison between couplings can be about, and Section 4.6.3 returns to the regime in which b^2 dominates.

Computational work is measured as elapsed execution time $W = BC$, with C the measured cost of one realization; each experiment states its timing protocol, and the cost of selecting a plan from preliminary runs is always reported separately from the cost of executing it.

Remark 2.1 (whole line versus window). The norm in (2.10) is a quadrature on the bounded window Ω and is used for every numerical comparison. The identity of Section 3.4 instead integrates over all of $\mathbb{R}$. We keep the two apart and never substitute one for the other.

3 Sign-Aware Coupling of Two Simulations

3.1 What the Mass Signs Do to an Antithetic Pair

Before specifying the coupling we isolate the calculation that dictates it. Consider one diffusion stage and condition on the state of both simulations immediately before it. Let a particle of simulation A sit at a with mass α , and let the particle matched to it in simulation B sit at b with mass β . After the stage their positions are

$$X_A = a + \sigma Z_A, \quad X_B = b + \sigma Z_B,$$

with $Z_A, Z_B \sim \mathcal{N}(0, 1)$; the coupling is the choice of their joint law. Abbreviate the two indicators by

$$\chi_A = \mathbf{1}\{X_A \leq x\}, \quad \chi_B = \mathbf{1}\{X_B \leq x\}.$$

By (2.3) the pair contributes $\frac{1}{2}(\alpha\chi_A + \beta\chi_B)$ to the pair mean at the point x . We construct the increment pairs independently across matched ranks. Under this restriction, contributions of distinct matched pairs are uncorrelated, and the conditional variance of the pair mean decomposes over pairs. For the pair above, the contribution is

$$\frac{1}{4} \left[\alpha^2 \text{Var} \chi_A + \beta^2 \text{Var} \chi_B + 2\alpha\beta \text{Cov}(\chi_A, \chi_B) \right]. \quad (3.1)$$

The first two terms depend only on the marginal laws of X_A and X_B , which every admissible coupling leaves alone. The coupling acts through the third term, and through it alone.

Write

$$p = \Phi\left(\frac{x-a}{\sigma}\right), \quad q = \Phi\left(\frac{x-b}{\sigma}\right)$$

for the two marginal probabilities, Φ being the standard normal distribution function. The covariance in (3.1) is $\text{Cov}(\chi_A, \chi_B) = \Pr\{X_A \leq x, X_B \leq x\} - pq$, and the joint probability is constrained by the classical Fréchet–Hoeffding bounds [17, Theorem 2.2.3 and Section 2.5],

$$\max(p+q-1, 0) \leq \Pr\{X_A \leq x, X_B \leq x\} \leq \min(p, q), \quad (3.2)$$

which hold for every coupling of the two marginals. Both bounds are attained here, and by the two coupling choices that are actually available. Taking $Z_B = -Z_A$ makes X_B a strictly decreasing function of X_A , so the pair is countermonotone and the left bound in (3.2) is an equality; taking $Z_B = Z_A$ makes them comonotone and the right bound is an equality. Writing

$$c_-(x) = \max(p+q-1, 0) - pq \leq 0, \quad c_+(x) = \min(p, q) - pq \geq 0, \quad (3.3)$$

reflection therefore realizes c_- and synchronization realizes c_+ , and no other coupling of the two increments can leave the covariance outside $[c_-(x), c_+(x)]$.

Restoring the mass product explains which endpoint we need. To minimize the pair's contribution to conditional variance, we make $\alpha\beta \text{Cov}$ as negative as possible. Reflection does this when $\alpha\beta > 0$, and synchronization does it when $\alpha\beta < 0$. We record this consequence of the classical bounds for the present reconstruction.

Proposition 3.1 (pairwise extremality of the switching rule). *Fix a diffusion stage, a matching, and the state of both simulations before the stage. For a matched pair with positions a, b and nonzero masses α, β , consider all joint laws of (Z_A, Z_B) with standard normal marginals. The contribution (3.1) of that pair to $\text{Var}(\bar{U}(x))$ is minimized, simultaneously for every $x \in \mathbb{R}$, by*

$$Z_B = \varepsilon Z_A, \quad \varepsilon = -\text{sgn}(\alpha) \text{sgn}(\beta),$$

and the minimum equals $\frac{1}{4}[\alpha^2 p(1-p) + \beta^2 q(1-q) + 2\alpha\beta c_{\mp}(x)]$ with c_- taken when $\alpha\beta > 0$ and c_+ when $\alpha\beta < 0$. Reflecting a pair with $\alpha\beta < 0$ attains the pointwise maximum instead.

Proof. The first two terms of (3.1) are functions of the marginals alone. By (3.2) the covariance lies in $[c_-(x), c_+(x)]$ for every coupling, and the two stated choices attain the endpoints because countermonotone and comonotone pairs realize the Fréchet–Hoeffding bounds. Multiplying by $2\alpha\beta$ reverses the ordering when $\alpha\beta < 0$, which gives the claim and its last sentence. $\square$

Proposition 3.1 explains how to choose the multiplier for a fixed matched pair. It does not settle the choice of matching, since a different matching changes which pairs exist, and the bounds (3.2) are then taken at different (a, b) . Nor does it extend along the horizon: the state entering a later stage depends on the coupling used at earlier stages, so a stage-by-stage optimum is not a terminal optimum. This division is the organizing principle of the paper. The multiplier is fixed by Proposition 3.1; the choice between matching by spatial rank and matching within sign classes is not settled by it, and is decided numerically in Section 4.

3.2 The Coupling and Two Alternatives

Both simulations start from the same deterministic initial state and are advanced in lockstep. At stage k let

$$S_k = (X^A, m^A, v^A, X^B, m^B, v^B)$$

denote the joint state after the sort and velocity update of Algorithm 1 and before the diffusion, and let $\mathcal{F}_k$ denote the information generated by the complete joint history up to that point, including all earlier random draws. A rule measurable with respect to $\mathcal{F}_k$ uses only what is known before the next draw is made.

Definition 3.2 (admissible coupling). An *admissible coupling* selects, at each stage k , a bijection $\pi_k : \{1, \dots, N\} \rightarrow \{1, \dots, N\}$ and multipliers $\varepsilon_k \in \{-1, +1\}^N$, both $\mathcal{F}_k$ -measurable, draws $Z^{(k)} \sim \mathcal{N}(0, I_N)$ independent of $\mathcal{F}_k$ and of all earlier draws, and sets

$$X_i^A \leftarrow X_i^A + \sigma Z_i^{(k)}, \quad X_j^B \leftarrow X_j^B + \sigma \varepsilon_{k,j} Z_{\pi_k(j)}^{(k)}. \quad (3.4)$$

Three admissible couplings are compared throughout. All give simulation A the same increments and differ only in (π_k, ε_k) .

REFLECT (reflected pairing). $\pi_k = \text{id}$ and $\varepsilon_{k,j} = -1$: match at equal spatial rank and reflect every pair. This is the direct transcription of antithetic sampling and serves as the common baseline.

SWITCH (sign-switched pairing). $\pi_k = \text{id}$ and

$$\varepsilon_{k,j} = -\text{sgn}(m_j^A) \text{sgn}(m_j^B), \quad (3.5)$$

the rule of Proposition 3.1: match at equal spatial rank, reflect a pair whose masses agree in sign, synchronize a pair whose masses differ.

WITHIN (within-sign pairing). π_k sends the r th particle of sign s in B to the r th particle of sign s in A , for $s \in \{+, -\}$, and $\varepsilon_{k,j} = -1$: restrict the matching so that no pair has unlike masses, then reflect every pair.

Algorithm 2: Coupled advance of two simulations.

Input: common initial state (X^0, m^0) sorted by position, u_- , steps K , step h , viscosity ν , flux derivative f' , policy $\in \{\text{REFLECT}, \text{SWITCH}, \text{WITHIN}\}$

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1  $(X^A, m^A) \leftarrow (X^0, m^0); (X^B, m^B) \leftarrow (X^0, m^0); \sigma \leftarrow \sqrt{2\nu h};$ 
2  $v_i^A \leftarrow f'(u_- + \sum_{j \leq i} m_j^A); v_i^B \leftarrow v_i^A;$ 
3 for  $k = 0, \dots, K-1$  do
4   foreach simulation  $Q \in \{A, B\}$  do
5      $X_i^Q \leftarrow X_i^Q + v_i^Q h;$  stable-sort  $(X^Q, m^Q)$  by  $X^Q$ ;
6      $v_i^Q \leftarrow f'(u_- + \sum_{j \leq i} m_j^Q);$ 
7   end
8   if policy = REFLECT then
9      $\pi \leftarrow \text{id}; \varepsilon_j \leftarrow -1;$ 
10  else if policy = SWITCH then
11     $\pi \leftarrow \text{id}; \varepsilon_j \leftarrow -\text{sgn}(m_j^A) \text{sgn}(m_j^B);$ 
12  else // WITHIN
13    foreach  $s \in \{+1, -1\}$  do
14       $I_s \leftarrow \{i : \text{sgn}(m_i^A) = s\}, J_s \leftarrow \{j : \text{sgn}(m_j^B) = s\},$  both increasing;
15      assert  $|I_s| = |J_s|;$ 
16       $\pi(J_s[r]) \leftarrow I_s[r]$  for  $r = 1, \dots, |I_s|;$   $\varepsilon \leftarrow -1$  on those  $j$ ;
17    end
18  end
19  draw  $Z \sim \mathcal{N}(0, I_N)$  independent of the joint history;
20   $X_i^A \leftarrow X_i^A + \sigma Z_i; X_j^B \leftarrow X_j^B + \sigma \varepsilon_j Z_{\pi(j)};$ 
21 end

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Output: $(X^A, m^A), (X^B, m^B)$, and the pair mean (2.8)

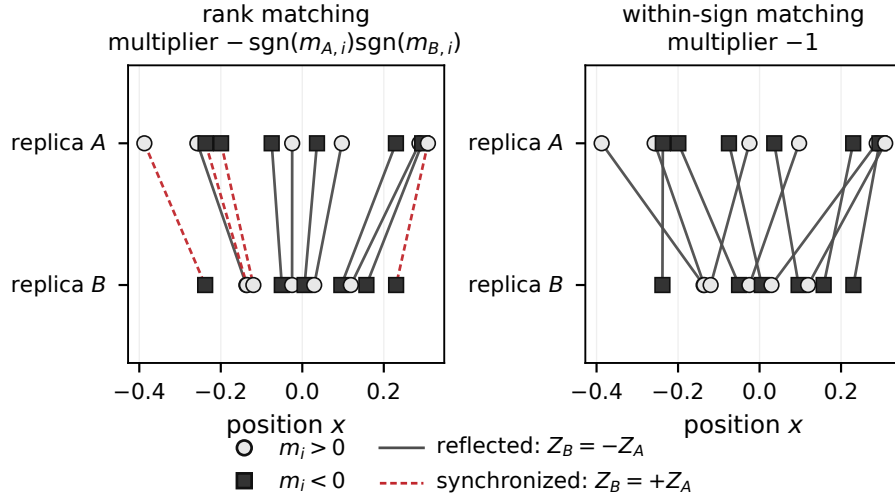


Fig. 1 The two matchings applied to the same paired state immediately before the tenth diffusion stage of the interleaved-sign configuration, with $N = 12$. The left panel pairs particles of equal spatial rank and synchronizes the increments of unlike-sign pairs. The right panel restricts partners to the same sign and reflects every pair. The state is identical in both panels, so the difference in separation comes only from the matching rule. Both rules preserve each simulation's law (Proposition 3.3).

By Proposition 3.1, REFLECT and SWITCH agree on every pair with like masses and disagree on every pair with unlike masses, where REFLECT makes precisely the worst available choice. WITHIN avoids unlike pairs altogether, at the price of matching particles that may be far apart; Figure 1 shows both matchings on one state. Algorithm 2 gives the coupled advance.

Four conditions in Definition 3.2 are needed for the law preservation of Section 3.3, and each can be violated by an implementation. The vector $Z^{(k)}$ must be freshly drawn and independent of the whole joint history; reusing one coordinate for two particles of the same simulation preserves each particle's marginal but changes the joint transition. The map π_k must be a bijection, so that no coordinate of $Z^{(k)}$ is consumed twice. The multipliers must have unit modulus, since any other value rescales the increment. Zero masses must be excluded or given a convention, because $\text{sgn}(0) = 0$ in (3.5) would suppress a particle's diffusion altogether; our initialization gives $|m_i| = \text{TV}(u_0)/N > 0$ and the implementation rejects a zero mass. Finally, WITHIN additionally requires equal sign-class counts in the two simulations, which the common deterministic initialization and the fixed masses guarantee.

3.3 Each Simulation Keeps its Law

The construction changes how the two simulations are related while preserving the law of each one. We can therefore retain results about a single simulation and its expectation; the variance of the coupled average is the quantity that requires a new comparison.

Proposition 3.3 (law preservation). *Let two simulations be advanced by Algorithm 1 from the same initial state under any admissible coupling. Then the law of the complete state trajectory $(X_k^A, m_k^A, v_k^A)_{k=0}^K$ of simulation A, and likewise that of simulation B, is exactly the law of an uncoupled single simulation. Only the joint law of the two is changed.*

The proof, in Appendix A, is an induction over stages: a predictable permutation of an i.i.d. Gaussian vector is i.i.d. Gaussian, and predictable ± 1 multipliers preserve $\mathcal{N}(0, \sigma^2)$ coordinatewise, so each simulation's one-step transition kernel is unchanged given the past. The same invariance of the Gaussian law under adapted orthogonal maps is used by Nüsken and Pavliotis [19, Lemma 74, Appendix A].

Remark 3.4 (what the proposition does and does not require). The argument uses the conditional Gaussian transition and the fact that all remaining operations of Algorithm 1 are deterministic functions of a single simulation's own state. It does not require the particle signs to be constant in time, nor the output functional to be linear in the particle measure. The present solver does keep signs fixed, but a scheme that updated masses would still preserve each simulation's law provided the updates are functions of that simulation's own state and the multipliers remain predictable and of unit modulus.

Corollary 3.5 (shared bias). *Under any admissible coupling, $\mathbb{E}\bar{U} = \mathbb{E}U^{\text{sing}}$ pointwise whenever these expectations exist. Hence the squared bias b^2 in (2.11) takes the same value for REFLECT, SWITCH and WITHIN, and equals that of the uncoupled solver.*

The corollary follows from Proposition 3.3 and linearity of the expectation across the two simulations, and it holds for the average of any integrable per-simulation functional. It does not hold for a nonlinear function of the pair mean, a distinction that becomes quantitative in Section 4.6. Corollary 3.5 also fixes the terms of every comparison below: since all three pairings share b^2 , they can be compared through V alone, and the planning rule of Section 4.5 may use one estimate of b^2 for all of them. What the corollary does not do is license any statement about the continuum: preserving a discrete law preserves whatever error that discrete law already has.

3.4 An Exact Identity for One Diffusion Stage

Proposition 3.1 compares couplings pointwise in x . Integrating that comparison over the line turns it into a single number, and the result is an exact expression for the variance that the switching rule removes at one stage. The following elementary identity supplies the integration.

Lemma 3.6 (kernel identity). *Fix $c \in \mathbb{R}$ and set $h_a(x) = \mathbf{1}\{x \geq a\} - \mathbf{1}\{x \geq c\}$, a function supported on the interval between a and c . For all $a, b \in \mathbb{R}$,*

$$\int_{\mathbb{R}} h_a(x) h_b(x) dx = \frac{1}{2}(|a - c| + |b - c| - |a - b|). \quad (3.6)$$

The verification is a case check on the order of a , b and c and is given in Appendix A. Writing the deviation of a reconstructed field from its far-field values in terms of these kernels lets the covariance of two reconstructions be expressed through the distances between their particles.

Proposition 3.7 (final-stage identity). *Let $g_s(d) = \mathbb{E}|d + sZ|$ with $Z \sim \mathcal{N}(0, 1)$. Condition on the paired state immediately before the last diffusion stage, let a_i and b_i be the positions of the particles matched at rank i , let $d_i = a_i - b_i$, and let m_i^A, m_i^B be their masses. Write*

$$V_{\bullet} = \int_{\mathbb{R}} \text{Var}(\bar{U}(x) \mid \mathcal{F}_{K-1}) dx$$

for the conditional whole-line integrated variance of the pair mean under coupling $\bullet$. Then

$$V_{\text{REFLECT}} - V_{\text{SWITCH}} = \frac{1}{4} \sum_{i: m_i^A m_i^B < 0} |m_i^A m_i^B| \left[g_{2\sigma}(d_i) - |d_i| \right] \geq 0. \quad (3.7)$$

Three features of (3.7) are worth reading off. Only pairs of unlike mass contribute, as Proposition 3.1 requires, since the two couplings agree elsewhere. Each contributing pair enters in proportion to the product of its mass magnitudes. And its contribution is governed by $g_{2\sigma}(d) - |d| \geq 0$, an increment that is largest when the matched separation $|d|$ is small compared with the diffusion scale σ and decays as $|d|/\sigma$ grows; the non-negativity is Jensen's inequality. Rank matching keeps $|d|$ small, which is one reason to prefer it, though not a proof that it wins overall.

Remark 3.8 (hypotheses and scope). The identity requires no condition on the total mass (2.4): the constant by which a reconstruction differs from its kernel representation is deterministic and cancels in the covariance, as the proof makes explicit. When $\sum_i m_i = 0$ the reference point c in (3.6) also drops out and one recovers the familiar product identity $\int_{\mathbb{R}} D_A D_B = -\frac{1}{2} \sum_{i,j} m_i^A m_j^B |X_i^A - X_j^B|$ for the centered reconstructions $D_{\bullet}$. Equation (3.7) is a statement about one stage, conditional on the state entering it; it does not assert that switching helps whenever signs disagree at an earlier stage, because the distribution of later states is itself coupling-dependent. It is a whole-line statement, and is never substituted for the windowed norm of Remark 2.1.

Two independent numerical checks of (3.7) are reported in Appendix A: a direct paired Monte Carlo over the final stage at six pre-final states of the two-pulse problem, and a quadrature of the joint indicator probabilities at a configuration with unequal mass magnitudes. Applied to the archived 64-seed runs, the identity accounts for 3.75%, 3.71% and 1.95% of the total measured reduction $V_{\text{REFLECT}} - V_{\text{SWITCH}}$ at three configurations, against measured single-stage shares of 4.23%, 5.23% and 2.91%. The exact result therefore explains a mechanism and a few percent of the effect; the remainder accumulates over the history and is measured, not proved.

3.5 When the Two Couplings Coincide

Lemma 3.9 (conditional identity). *Start REFLECT and SWITCH from the same state and drive them with the same sequence of draws. Their complete states coincide at every stage up to and including the first stage at which some matched pair has unlike masses. If no such stage occurs before K , the two runs are identical.*

The proof is immediate from (3.5): while every matched pair has like masses, $\varepsilon_{k,j} = -1$, which is REFLECT.

Corollary 3.10 (one-sign case). *If all gradient masses share one sign, the hypothesis of Lemma 3.9 holds for every history, and SWITCH and REFLECT are the same algorithm.*

Corollary 3.10 is a genuine limitation rather than a curiosity: for monotone initial data the construction does exactly nothing, and no experiment can show otherwise. We verified it on the complete state, positions, masses and carried

velocities, for 200 seeds under both heat and Burgers dynamics, with maximum absolute difference exactly zero in every seed.

It is tempting to expect the same conclusion when the positive and negative regions merely start far apart, since the rank order then groups particles by sign. That expectation is false, and the reason is elementary: Gaussian support is unbounded, so for two particles separated by d the probability that exactly one of the two simulations reverses their order is $\Pr\{|Z_1 - Z_2| > d/\sigma\} > 0$ at every finite d . Once an order reverses, a matched pair of unlike mass appears and the two couplings separate. On the spatially separated configuration of Section 4.7 at $N = 2048$ with 200 fresh seeds, the two couplings diverge in $58/200 = 0.290$ $[0.228, 0.358]$ of seeds under heat dynamics and $59/200 = 0.295$ $[0.233, 0.363]$ under Burgers, with median first divergence at step 187 of 200.

3.6 Two Routes That Are Closed

The pairwise result of Section 3.1 is exact but local, and the identity of Section 3.4 is exact but confined to one stage. It is natural to ask what blocks an unconditional terminal statement. Two obvious routes are closed, and saying so is more useful than leaving them open.

Remark 3.11 (coordinatewise monotonicity fails). Antithetic sampling is guaranteed to help when the response is monotone in each input, so one might hope to establish that the terminal continuation value is monotone in each particle position, with an orientation set by the mass sign. It is not. In a two-particle, transport-dominated configuration ($m = \pm \frac{1}{2}$, $h = 0.005$, $\nu = 10^{-10}$, hence $\sigma = 10^{-6}$ against a transport displacement of 10^{-2}), the conditional expectation $\mathbb{E}[U(x, T) \mid \text{state}]$ at a probe point *increases* by 0.5 when a positive-mass particle is displaced to the right, the opposite of the required orientation. The configuration was traced against the solver over 512 seeds, every seed reproducing the deterministic outcome, and the probability of an exceptional draw is bounded analytically by $\exp(-5003)$. Appendix B gives the trace.

The counterexample rules out the unrestricted argument. It does not show that SWITCH is ever worse than REFLECT, and the configuration is far outside the diffusion-dominated regime: its ratio of mass-dependent transport to diffusion scale is $R = |m|h/\sigma = 2500$. Whether a terminal comparison can be proved under a small- R assumption remains open.

Remark 3.12 (sorted matching is optimal for absolute distance). The second route would compare the two matchings geometrically. For two equally sized finite sets of reals listed in increasing order, the identity bijection minimizes $\sum_i |a_i - b_{\pi(i)}|$ over all bijections π ; this is the monotone rearrangement, which is the optimal transport map on the line for any convex cost [26]. Rank matching *is* that bijection, and within-sign matching is a bijection constrained to respect the sign classes, so

$$\sum_i |a_i - b_i| \leq \sum_i |a_i - b_{\pi_{\text{WITHIN}}(i)}| \quad (3.8)$$

with probability one.

Inequality (3.8) tells us that rank matching always produces the closer partners, and by Proposition 3.7 close partners are what the final-stage gain rewards. It does not tell us that rank matching wins, because the two rules differ in which pairs exist and therefore in which mass products multiply those distances. In particular, any measured ratio of matched separations is bounded above by 1 by construction, so such a ratio reports the size of the geometric penalty imposed by the sign-class constraint and cannot by itself identify the cause of a variance difference. We report it in Section 4.4 on that understanding.

4 Numerical Results

Section 3 fixes the multiplier and leaves the matching open. The experiments below therefore lead with the comparison between SWITCH and WITHIN, the two sign-aware rules, and use REFLECT as the common baseline against which both are measured. Six tests are reported: the benchmark comparison of the three pairings; a comparison with two standard variance-reduction techniques applied to the same solver; a variation of the initial profile, the observation time and the flux; execution time at prescribed accuracy with plans frozen before evaluation; nonlinear functionals and the bias-dominated regime; and a set of signed-structure controls that locate the configurations in which the construction does nothing.

4.1 Test Configuration and Statistical Conventions

Unless stated otherwise the test problem is the asymmetric signed two-pulse initial condition

$$u_0(x) = 0.8 e^{-(x+0.7)^2/(2 \cdot 0.5^2)} - 0.5 e^{-(x-0.4)^2/(2 \cdot 0.3^2)}, \quad (4.1)$$

with $f(u) = u^2/2$, $\nu = 0.1$, $T = 1$ and $h = 0.005$, hence $K = 200$ steps, reconstructed on $\Omega = [-5, 5]$ at $L = 400$ points. Its derivative has three resolved sign regions, so both sign classes are populated and the classes interleave once; unlike-sign matched pairs therefore occur, which is the situation the coupling addresses. Reference solutions are Fourier pseudospectral on $[-16, 16]$ with $M = 2048$ modes, $\frac{2}{3}$ dealiasing and an integrating-factor RK4 step of 10^{-4} . For Burgers this agrees with a stabilized whole-line Cole–Hopf evaluation to 5.6×10^{-14} in maximum norm, and for the cubic flux it self-converges to 1.6×10^{-13} ; both are negligible beside the field errors reported below, which are of order 10^{-6} . Appendix C gives the initializations and reference calculations.

Every experiment uses recorded deterministic seed keys, and independent replicates use distinct streams. Whether two comparison arms share a stream is recorded with each experiment, because it determines the resampling scheme: bootstrap intervals resample complete realizations [9], and arms that share a stream are resampled together so that their dependence is preserved. The

main field, control and nonlinear-error tables use 6000 percentile-bootstrap resamples; the profile/time and structure tables retain their archived 3000 and 2000. Variance ratios of scalar station observables use $F(n-1, n-1)$ intervals for independent arms. Held-out errors use t intervals on 32 independent block errors, with a Bonferroni correction across the arms of each experiment. Divergence frequencies use exact binomial intervals. Appendix E reports a coverage calibration of the two scalar interval procedures and states its limits.

Throughout, a ratio below 1 favors the numerator, and an interval containing 1 means the comparison is unresolved at the sample size used. Such an interval records the absence of evidence, not evidence of absence.

4.2 Test 1: the Three Pairings on the Two-Pulse Problem

The first test compares the three pairings at $N = 8192$ over 48 replicates. Table 1 reports the integrated variance and integrated mean squared error of the pair mean. Restricting the matching to sign classes is already most of what is available: WITHIN reduces the integrated variance to 0.172 of the reflected value. Retaining the spatial matching and switching the multiplier reduces it further, to 0.127 of reflected pairing and 0.739 of the within-sign value, and the interval for that further reduction excludes 1.

Table 1 Integrated variance and integrated mean squared error over $\Omega = [-5, 5]$ for the pair mean, two-pulse Burgers, $N = 8192$, $h = 0.005$, 48 replicates, with whole-replicate percentile bootstrap intervals. The absolute integrated variances are 4.326×10^{-5} (reflected), 7.433×10^{-6} (within-sign), 5.492×10^{-6} (sign-switched).

comparison	integrated variance ratio	integrated MSE ratio
within-sign / reflected	0.1718 [0.1450, 0.2038]	0.2056 [0.1738, 0.2443]
sign-switched / reflected	0.1269 [0.1073, 0.1511]	0.1684 [0.1425, 0.2007]
sign-switched / within-sign	0.7389 [0.6817, 0.8006]	0.8193 [0.7542, 0.8928]

Figure 2 shows how the reduction arises and where it sits. The upper panels display, for one replicate, the pointwise error of each simulation and of their mean. Under REFLECT the two errors are only partly opposed and their mean retains a visible signature near the steep core of the profile; under SWITCH they are close to mirror images and the mean is much flatter. The lower panel gives the ratio of pointwise variances: the reduction against REFLECT is concentrated where the reference gradient is large and is weak on the flat flanks, while the reduction against WITHIN is smaller and more evenly spread. The integrated ratio is not representative of a single location; its value of 0.117 at this configuration should be read against a median pointwise ratio of 0.495, the difference being that the integral is dominated by the high-variance region, whose quintile carries 65% of the total variance and has median pointwise ratio 0.095. Section 4.6 quantifies what a reader whose quantity of interest lives elsewhere should expect.

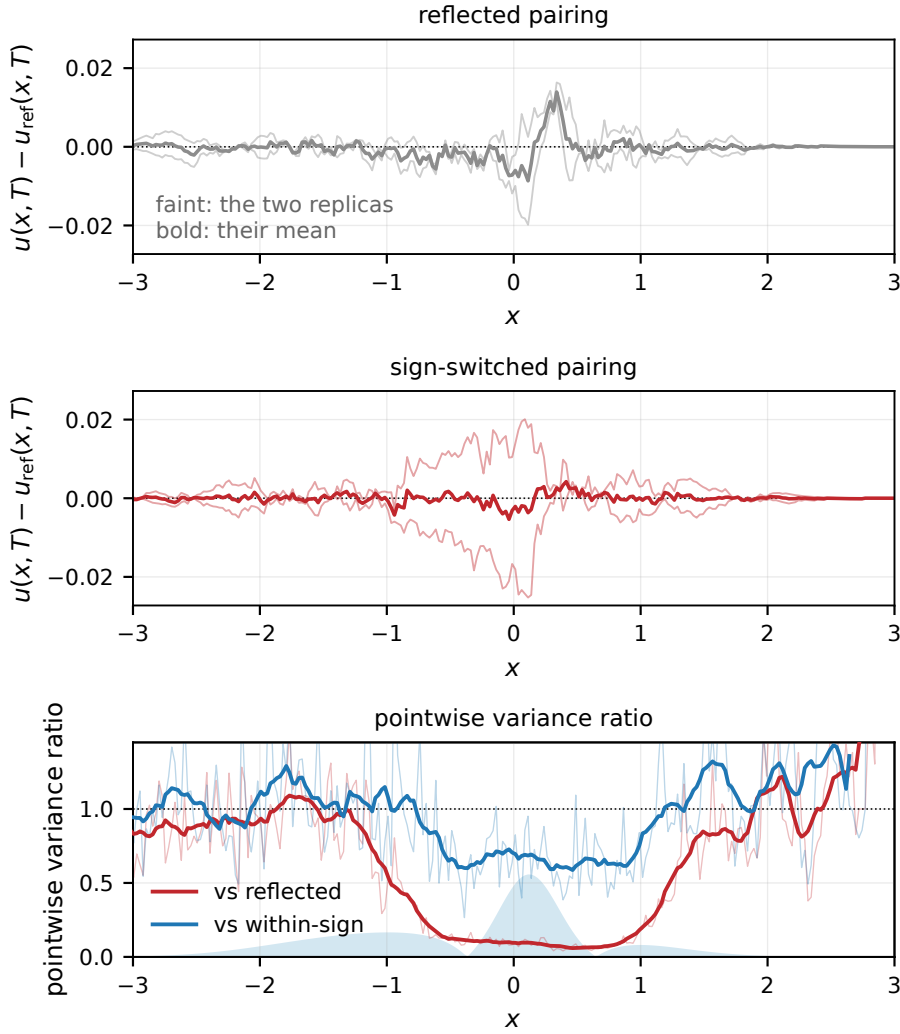


Fig. 2 Two-pulse Burgers, $N = 8192$, 48 replicates, from the independently replayed archive. The top and middle panels show the pointwise error of each replica (faint) and of their mean (bold) on one replicate. The bottom panel shows the ratio of pointwise variance of the sign-switched pair mean to that of the reflected (red) and within-sign (blue) pair means. Thin lines are the raw ratios over 48 replicates and thick lines a 13-point running mean. The shaded profile is $|\partial_x u_{\text{ref}}|$, scaled to the panel.

Cost must be counted before the reduction can be interpreted. A pair costs 1.93 times a single trajectory at $N = 8192$, 0.1950 s against 0.1009 s in serial execution. Among the three pairings, one replicate costs 0.1941 s for REFLECT, 0.1995 s for SWITCH and 0.2117 s for WITHIN, so the switch costs 1.028 times reflected pairing and 0.943 times within-sign pairing; at $N = 2048$

the corresponding figures are 1.040 and 0.908. Combining the measured cost with Table 1 gives a variance-times-cost ratio of 0.697 against within-sign pairing. These are implementation measurements on one serial machine. In these experiments the sign classes always have equal counts in both simulations, so the handling of unequal counts cannot be what makes WITHIN slower here; partitioning and gathering are the plausible causes, but we did not profile them, and we therefore claim no intrinsic cost advantage over every within-sign implementation. Parallel execution, in which the two simulations must exchange matched masses at each stage, is not addressed.

4.3 Test 2: Conditional Expectation and Control Variates

A reader with a variance problem need not change the coupling at all. The second test therefore applies two standard techniques to the same solver [11, Chapter 4]: conditional expectation, which integrates the final Gaussian diffusion analytically instead of sampling it, and control variates with known Gaussian means, one built from each particle’s accumulated Brownian displacement and one that also retains its initial transport velocity. Coefficients are fitted on a disjoint training sample and then held fixed. Appendix D gives the estimators and the training rule. All arms share seed streams, and the intervals in Table 2 retain that pairing.

Table 2 Alternatives on the same solver. Integrated variance relative to reflected pairing, with paired whole-replicate bootstrap intervals, at $N = 1600$, $h = 0.005$, 64 seeds. “ $v \times c$ ” multiplies the variance ratio by the measured per-replicate cost ratio. One-off coefficient training, where required, is listed separately and is not amortized in $v \times c$. c^* denotes a coefficient fitted on a disjoint batch and frozen.

arm	variance / reflected	$v \times c$	training (s)	variance / within-sign
Burgers flux				
reflected	1.0000	1.000	–	–
+ conditional final stage	0.8813 [0.852, 0.909]	1.612	–	–
+ heat control, c^*	0.8440 [0.821, 0.869]	1.170	0.488	–
+ frozen-velocity control, c^*	0.6724 [0.617, 0.740]	0.923	0.466	–
within-sign	0.2083 [0.179, 0.245]	0.243	–	1
sign-switched	0.1649 [0.142, 0.194]	0.174	–	0.7916 [0.762, 0.825]
+ frozen-velocity control, c^*	0.1642 [0.141, 0.194]	0.237	0.490	0.7886 [0.758, 0.822]
+ conditional final stage	0.0804 [0.069, 0.095]	0.151	–	0.3861 [0.362, 0.412]
Cubic flux				
reflected	1.0000	1.000	–	–
+ frozen-velocity control, c^*	0.5031 [0.458, 0.552]	0.697	0.448	–
within-sign	0.2263 [0.196, 0.262]	0.268	–	1
sign-switched	0.1739 [0.151, 0.201]	0.184	–	0.7688 [0.741, 0.797]
+ heat control, c^*	0.1684 [0.146, 0.194]	0.248	0.496	0.7442 [0.716, 0.773]
+ conditional final stage	0.0818 [0.070, 0.096]	0.158	–	0.3614 [0.340, 0.383]

Three readings of Table 2 matter. First, the best control applied to reflected pairing reaches relative variance 0.672 on Burgers but variance-times-cost 0.923, before 0.47 s of coefficient training; at this configuration changing the coupling is worth more than adding a good control to the unfavorable coupling. Second, once the coupling is switched, these controls add little: the fitted frozen-velocity control moves the relative variance from 0.1649 to 0.1642 on Burgers, and on the cubic flux the heat control moves it from 0.1739 to 0.1684, a resolved ratio of 0.968. This is a statement about these two controls at these configurations, not a claim that no control can help. Third, conditional expectation is complementary rather than competing: applied on top of the switched coupling it reduces relative variance from 0.1649 to 0.0804 and gives the smallest variance-times-cost in the table, 0.151. Its benefit depends on how many conditional evaluations the output requires; raising that number from 201 to 1601 moves the variance-cost ratio of adding the conditional stage from 0.614 to 1.485. We therefore report the coupling and the conditional stage separately, so that neither inflates the other.

A fourth alternative, a randomized quasi-Monte Carlo adaptation of the diffusion construction of Lécot [15], is an arm of Test 4 in Section 4.5, where it attains the common target with 46 trajectories against four.

4.4 Test 3: Initial Profile, Observation Time, and Flux

A single configuration establishes little about a method. The third test varies three things that should matter: the sign structure of the initial profile, the observation time, and the flux. Besides the two-pulse profile (4.1) we use

$$\begin{aligned} u_{0,\text{bump}}(x) &= 0.8 \exp\left(-\frac{(x+0.2)^2}{2 \cdot 0.45^2}\right), \\ u_{0,\text{osc}}(x) &= 0.45 \sin(3x) \exp\left(-\frac{x^2}{2 \cdot 1.1^2}\right), \end{aligned} \quad (4.2)$$

whose derivatives have 2 and 13 resolved sign regions against the two-pulse profile's 3; the numerical threshold defining the count is given in Appendix C. Each profile is observed at $T = 0.25, 1.0$ and 2.0 , with $N = 2048$ and 48 replicates.

Table 3 Integrated variance ratio of sign-switched to within-sign pairing, $N = 2048$, $h = 0.005$, 48 replicates, whole-replicate bootstrap intervals.

profile	sign regions	$T = 0.25$	$T = 1.0$	$T = 2.0$
single bump	2	0.958 [0.886, 1.039]	0.790 [0.727, 0.862]	0.810 [0.735, 0.893]
two-pulse	3	0.895 [0.830, 0.967]	0.785 [0.713, 0.866]	0.793 [0.702, 0.889]
oscillatory	13	0.932 [0.876, 0.990]	0.667 [0.613, 0.725]	0.595 [0.544, 0.655]

Eight of the nine configurations in Table 3 have intervals below 1. The exception is the single bump at the earliest time, whose interval [0.886, 1.039]

leaves the comparison unresolved. All three profiles show a larger reduction at $T = 1$ and $T = 2$ than at $T = 0.25$, and the oscillatory profile shows the largest reduction at the two later times. The reduction does not grow monotonically with time: between $T = 1$ and $T = 2$ the ratio rises from 0.790 to 0.810 for the single bump and from 0.785 to 0.793 for the two-pulse profile.

It is tempting to read the ordering of Table 3 as an effect of sign-region count, since that is the quantity we set out to vary. The three profiles also differ in amplitude, shape and gradient magnitude, so the ordering does not isolate that cause. We therefore examine the matching geometry separately, by applying both matching rules to the same paired state so that the history is held fixed.

Table 4 Mean matched-pair separation under rank matching divided by that under within-sign matching, both rules applied to the same paired state so the history is held fixed, with $N = 2048$, 24 replicates, ± 2 standard errors. By Remark 3.12 every entry is at most 1 with probability one.

profile	sign regions	$T = 0.25$	$T = 1.0$	$T = 2.0$
single bump	2	0.928 [0.908, 0.948]	0.851 [0.817, 0.885]	0.790 [0.741, 0.840]
two-pulse	3	0.897 [0.879, 0.916]	0.839 [0.803, 0.875]	0.820 [0.762, 0.878]
oscillatory	13	0.847 [0.832, 0.862]	0.693 [0.655, 0.732]	0.676 [0.624, 0.728]

Table 4 measures how much extra distance the within-sign constraint imposes. Its entries are at most 1 by Remark 3.12, and they are smallest for the oscillatory profile at the later times, which is where the variance reduction is largest. The two diagnostics do not, however, rank all three profiles alike: at $T = 2$ the separation ratios order them oscillatory, single bump, two-pulse, whereas the variance ratios order them oscillatory, two-pulse, single bump. The geometric comparison therefore supports the interpretation that nearby partners are worth retaining, and is consistent with Proposition 3.7, but it does not establish that separation causes the full-history reduction.

Finally we change the flux. Replacing $f(u) = u^2/2$ by $f(u) = u^3/3$ gives

$$f'(u) = u^2 \geq 0,$$

so advection is rightward or stationary rather than bidirectional, and the characteristic speed depends quadratically on the field. With $N = 1600$, $h = 0.005$ and the same 64 seeds, the ratio of SWITCH to WITHIN integrated variance is 0.7916 [0.762, 0.825] for Burgers and 0.7688 [0.741, 0.797] for the cubic flux (Table 2). The reduction does not require transport in both directions. Two fluxes are two fluxes, and we do not extrapolate to general f .

4.5 Test 4: Execution Time at a Prescribed Accuracy

Variance ratios at fixed settings are not work. The fourth test measures execution time at a specified expected error, with the estimator plan fixed before the

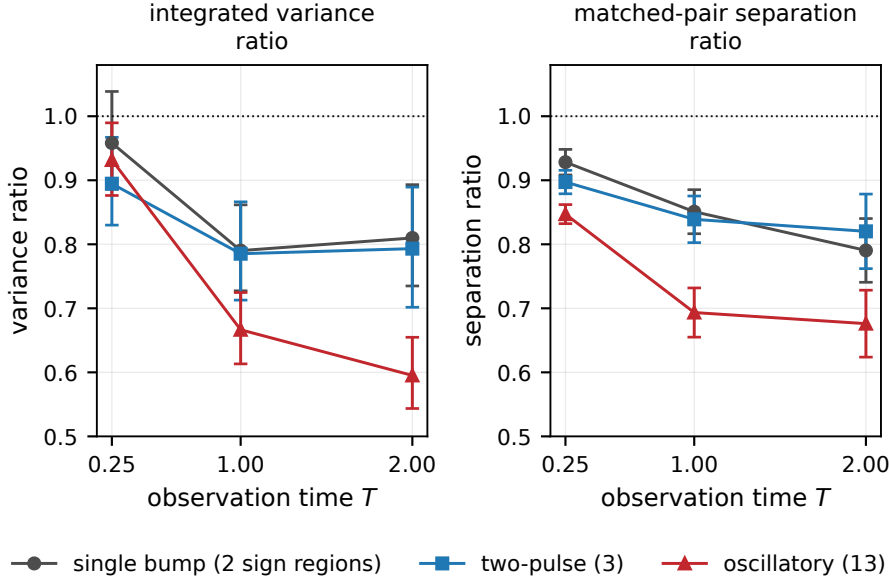


Fig. 3 The left panel shows the integrated variance ratio of sign-switched to within-sign pairing across three fixed profiles and three observation times (Table 3). The right panel shows the matched-pair separation ratio at a fixed paired state (Table 4), which cannot exceed 1 by Remark 3.12. It measures the increase in distance from constraining the matching to respect sign classes, not the variance mechanism.

evaluation data are seen. For each arm a preliminary run supplied estimates of the squared bias, the integrated variance, and the cost per replicate; the replicate count was then set to the smallest positive integer B with

$$b_{\text{hi}}^2 + \frac{V_{\text{hi}}}{B} \leq \frac{\text{tol}}{\kappa}, \quad \kappa = 1.3, \quad (4.3)$$

using upper interval endpoints and the headroom factor κ ; and the frozen plan was executed in 32 independent blocks on an unused seed stream, a block being one complete B -replicate estimate. By Corollary 3.5 the three pairings and the unpaired ensemble share a bias, so one pilot estimate of b^2 , taken from the switched arm, serves all of them; the quasi-Monte Carlo arm does not preserve the baseline law and uses its own. No estimate is truncated at zero: the squared bias is estimated as $\|\hat{U} - u_{\text{ref}}\|_{\Omega}^2 - V/B_{\text{pilot}}$, which is unbiased and may take small negative values, and it is the upper endpoint b_{hi}^2 that enters (4.3).

The two experiments differ in target, particle count and comparators, and we report them separately rather than pooling them. Experiment A was designed before within-sign pairing entered the study and does not contain it; it compares the switched coupling with an unpaired ensemble, with reflected pairing, and with the quasi-Monte Carlo adaptation. Experiment B compares the three pairings.

4.5.1 Experiment A: Unpaired and Quasi-Monte Carlo Comparators

Experiment A uses $N = 8192$ and targets an expected integrated squared error of 7×10^{-6} , with pilot key 1601 and evaluation key 1702. The target was chosen from the pilot and is therefore pilot-informed rather than independent of the data.

The quasi-Monte Carlo arm adapts Lécot’s rank-and-increment map [15, Eq. (45)] to our transport update. One scrambled Sobol’ sequence is generated per replicate, and successive time steps consume successive blocks of N points; the first coordinate assigns a rank by $\lfloor Np_1 \rfloor$ and the second supplies the increment $\sigma\Phi^{-1}(p_2)$. This is an adaptation and not a reproduction of the published method, which uses a different transport rule and a Faure sequence; its interacting law differs from the baseline, so its bias is estimated separately and its total error is what the comparison uses. Appendix D records the engine settings, the scrambling, and the endpoint handling.

Table 5 Held-out experiment A. Frozen plans, unused evaluation key, 32 blocks, $N = 8192$, target 7×10^{-6} . “Trajectories” counts solver runs, which is $2B$ for a paired arm and B otherwise. The upper bound is a Bonferroni-corrected 95% t bound across the four arms ($t_{31, 1-0.05/8} = 2.652$). Every arm’s bound lies below the target.

arm	B	trajectories	attained error	simultaneous upper bound	work (s)
unpaired ensemble	65	65	5.347×10^{-6}	6.743×10^{-6}	6.658
reflected pairing	17	34	5.031×10^{-6}	6.362×10^{-6}	3.340
RQMC adaptation	46	46	4.024×10^{-6}	4.873×10^{-6}	6.848
sign-switched pairing	2	4	5.100×10^{-6}	5.611×10^{-6}	0.407

4.5.2 Experiment B: Within-Sign Pairing

Experiment B uses $N = 2048$ and targets 2.5×10^{-5} , with pilot key 2310, evaluation key 2320, and a tuning cost of 4.05 s reported separately from execution. It uses the standard two-pulse configuration rather than the oscillatory profile, which would favor the switched coupling.

Table 6 Held-out experiment B. Frozen plans, unused evaluation key, 32 blocks, $N = 2048$, target 2.5×10^{-5} , Bonferroni bound across three arms ($t_{31, 1-0.05/6} = 2.531$). Work is end-to-end with interleaved arm order.

arm	B	trajectories	attained error	simultaneous upper bound	work (s)
reflected pairing	18	36	1.404×10^{-5}	1.830×10^{-5}	0.615
within-sign pairing	3	6	1.769×10^{-5}	1.936×10^{-5}	0.119
sign-switched pairing	2	4	1.976×10^{-5}	2.203×10^{-5}	0.0715

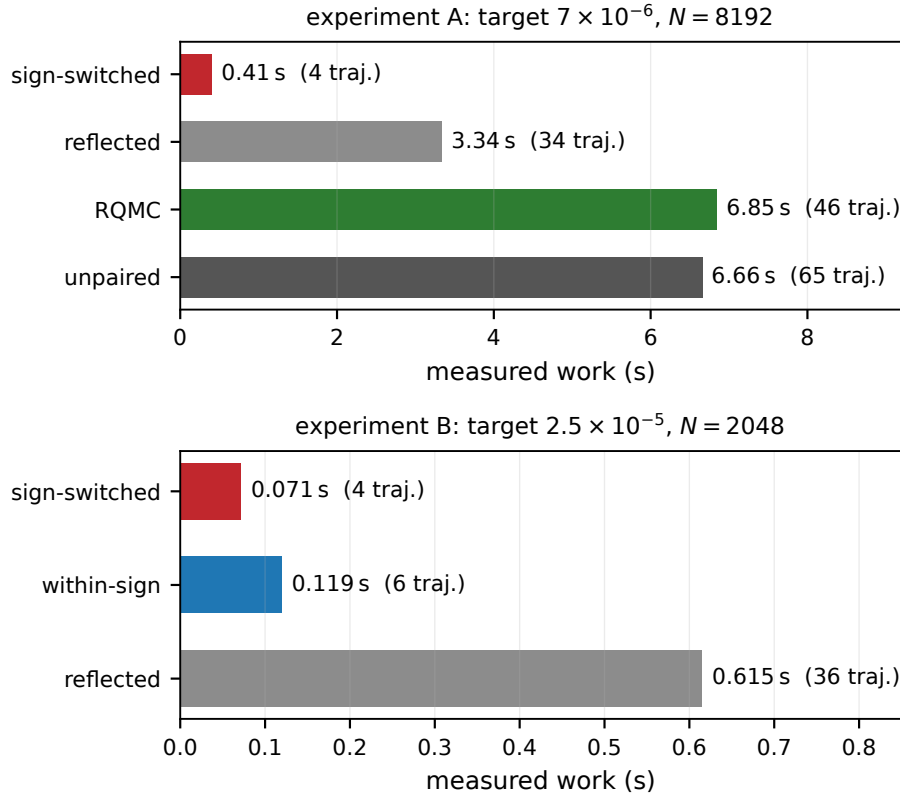


Fig. 4 Measured work to attain a common expected-error target in the two held-out experiments, which have different targets, different N and different arms and are not to be pooled. The top panel reports experiment A (Table 5), and the bottom reports experiment B (Table 6).

All arms of both experiments attain their target, pointwise and simultaneously. In Experiment B the measured work ratio is 0.599 against within-sign pairing and 0.116 against reflected pairing. Three qualifications belong with those numbers. The arms do not land at equal attained error, reflected pairing overshooting the target by the largest margin, so this is attainment by a selected plan and not an equal-error comparison. At $B = 2$ against $B = 3$ the integer granularity carries much of the ratio: $\frac{2}{3}$ times the measured cost ratio 0.908 is 0.605, close to the observed 0.599, so the experiment mainly establishes that the switched arm needs one fewer replicate here. And the plans were selected from a pilot rather than optimized. What the experiment does establish is that the variance reduction survives translation into execution time on an unused seed stream, which a fixed-setting variance ratio cannot show.

4.6 Test 5: Observables and the Bias-Dominated Regime

4.6.1 Dependence on Position

At $N = 8192$ with 48 replicates on a separate seed key, the variance ratio of SWITCH to REFLECT at four reconstruction points is 0.636 [0.357, 1.135] at the left flank $x = -2.0$, 0.143 [0.080, 0.255] at the left pulse $x = -0.7$, 0.093 [0.052, 0.166] at the steepest core $x = +0.1$, and 0.735 [0.412, 1.311] at the right flank $x = +1.5$, with $F(47, 47)$ intervals. The two flank comparisons are unresolved. The reduction is a property of the region where the field varies, which is also where the particles are.

4.6.2 Nonlinear Functionals

For a nonlinear functional Q the two natural estimators are not the same object. Applying Q to the pair mean gives $Q(\bar{U})$, whereas averaging the functional over the two simulations gives $\frac{1}{2}(Q(U^{(A)}) + Q(U^{(B)}))$. By Proposition 3.3 the second has exactly the expectation of Q evaluated on a single uncoupled simulation, so its bias is the single-simulation bias whatever the coupling; the first enjoys no such guarantee, because Corollary 3.5 does not survive a nonlinear map. A variance ratio is therefore not an accuracy ratio, and we report the empirical mean squared error against the reference,

$$\widehat{\text{MSE}} = \frac{1}{n} \sum_{r=1}^n (Q_r - Q_{\text{ref}})^2 = \widehat{\text{bias}}^2 + \frac{n-1}{n} s^2, \quad (4.4)$$

rather than the plug-in $\widehat{\text{bias}}^2 + s^2$, which exceeds (4.4) by s^2/n , here 0.7% to 2.0%. Both functionals are rule-dependent and we state the rules: $\max|\partial_x u|$ uses central differences of the reconstruction on the output grid, and the level crossing is the first downward crossing of $u = 0.25$ to the right of the global maximum, by linear interpolation, with replicates lacking such a crossing recorded as censored. None occurred in any cell.

Table 7 Nonlinear observables at $N = 8192$, 48 replicates, against the spectral reference, from the independently replayed archive. Ratios are of the empirical mean squared error (4.4), with whole-replicate bootstrap intervals. The last column is the squared-bias share of the sign-switched arm’s empirical MSE. An interval containing 1 is unresolved.

functional	estimator	switched / reflected	switched / within-sign	bias ² share
$\max \partial_x u $	Q of the mean	0.416 [0.244, 0.702]	0.602 [0.359, 0.984]	0.56
	mean of Q	0.832 [0.603, 1.139]	1.058 [0.803, 1.394]	0.87
level $x(u = 0.25)$	Q of the mean	0.129 [0.070, 0.247]	0.855 [0.497, 1.419]	0.19
	mean of Q	0.090 [0.053, 0.158]	0.847 [0.497, 1.486]	0.30

Table 7 contains the clearest applicability boundary in the paper. For the level crossing the switched coupling reduces total error by roughly an order

of magnitude against reflected pairing under either estimator. For the peak gradient it does not: with the mean-of- Q estimator the ratio against reflected pairing is 0.832 [0.603, 1.139] and against within-sign pairing 1.058 [0.803, 1.394], both unresolved, because 87% of that estimator’s error is squared bias, which Corollary 3.5 makes common to all three couplings. A variance reduction cannot improve an error that is mostly bias. The corresponding variance ratios against reflected pairing are 0.410 for Q of the mean and 0.686 for the mean of Q , factors of 2.4 and 1.5 rather than an order of magnitude.

4.6.3 The Bias-Dominated Regime

The same arithmetic governs the field itself. Since the couplings share b^2 by Corollary 3.5, (2.11) gives

$$\frac{\mathcal{E}_{\text{SWITCH}}(B)}{\mathcal{E}_{\text{REFLECT}}(B)} = \frac{b^2 + V_{\text{SWITCH}}/B}{b^2 + V_{\text{REFLECT}}/B} \longrightarrow 1 \quad \text{as } B \rightarrow \infty, \quad (4.5)$$

so at fixed N and h any coupling advantage is a small- B phenomenon. The scale on which it fades is $B^* = V/b^2$, at which the sampling and squared-bias contributions are equal; for the switched arm this is 10.3 at $N = 6400$, $h = 0.0025$, 4.5 at $N = 6400$, $h = 0.005$, and 35.3 at $N = 1600$, $h = 0.0025$.

Comparing at equal B is not the same as comparing at equal expense, since the two arms have different per-replicate costs C_{SWITCH} and C_{REFLECT} . Setting $B_\bullet = W/C_\bullet$ for a common work budget W gives the quantity plotted in Figure 5,

$$\frac{b^2 + V_{\text{SWITCH}}C_{\text{SWITCH}}/W}{b^2 + V_{\text{REFLECT}}C_{\text{REFLECT}}/W}, \quad (4.6)$$

whose limit as $W \rightarrow \infty$ is again 1. The crossover scale becomes $W = C_{\text{SWITCH}}V_{\text{SWITCH}}/b^2$, marked by the circles in the figure. It is a crossover and not a cutoff: there the ratio is still 0.18 to 0.26, and it approaches 1 only some two decades of work later.

The practical reading is the one a user can act on. Coupling buys accuracy while sampling error still contributes materially to the requested quantity, which is the regime in which one is paying for solver trajectories. Once discretization error dominates, the remedy is to refine N or h , and no coupling will substitute for that.

4.7 Test 6: Where the Construction Does Nothing

The last test maps the effect against the signed structure of the initial data at $N = 2048$, using four configurations: alternating signs on a short interval, two blocks of opposite sign driven together, two blocks held far apart, and a single sign. Their definitions are in Appendix C. Each is run under both heat and Burgers dynamics, the heat case isolating the coupling from transport.

The rows of Table 8 are informative in different ways and should not be read alike. The interleaved configuration is a cancellation test rather than

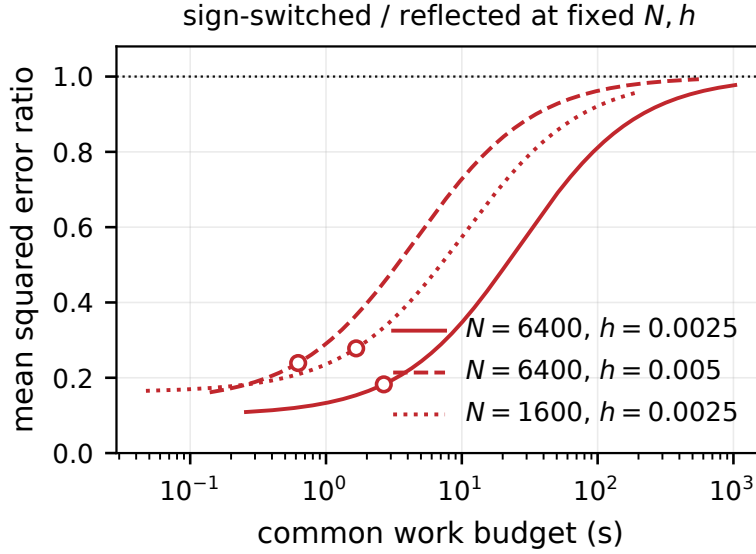


Fig. 5 Model mean squared error ratio (4.6) at equal work, at three archived (N, h) cells of the two-pulse problem with 64 paired seeds. The shared squared bias b^2 is the same estimate in numerator and denominator, as Corollary 3.5 requires. Circles mark $W = C_{\text{SWITCH}} V_{\text{SWITCH}} / b^2$. Three configurations do not establish a convergence rate or independently resolve the effects of N and h .

Table 8 Integrated variance ratio of sign-switched to reflected pairing by signed structure, $N = 2048$, $h = 0.005$, $T = 1$, 48 replicates, whole-replicate bootstrap intervals. The two arms of each cell were run on different seed streams, so these are ratios of independent estimates. The one-sign row is therefore an estimate of a quantity that Corollary 3.10 makes exactly 1, and its departure from 1 is sampling noise. Exact identity is established separately, by a matched-seed full-state comparison.

structure	dynamics	variance ratio	reading
interleaved (alternating signs)	heat	0.049 [0.040, 0.061]	variance reduced
	Burgers	0.063 [0.052, 0.077]	variance reduced
merging blocks	heat	0.476 [0.421, 0.545]	variance reduced
	Burgers	0.481 [0.426, 0.544]	variance reduced
separated blocks	heat	0.994 [0.917, 1.076]	unresolved
	Burgers	1.013 [0.930, 1.101]	unresolved
one sign	heat	0.987 [0.902, 1.078]	exactly 1 by Cor. 3.10
	Burgers	0.967 [0.822, 1.174]	exactly 1 by Cor. 3.10

a refinement study: its alternating masses $\pm 1/N$ on a fixed support give a reconstruction whose amplitude falls like $1/N$, from 2.5×10^{-3} at $N = 400$ to 1.2×10^{-4} at $N = 8192$, while the total absolute mass stays at 1. Increasing N there changes the represented field and not merely its resolution, so its ratios cannot be read as convergence under refinement. The merging and separated configurations hold the amplitude at 0.5 and are refinements of one fixed profile.

The unresolved rows for separated blocks are consistent with the divergence analysis of Section 3.5: the two couplings agree on about 71% of seeds, and

when they do diverge they do so at a median step of 187 out of 200, leaving little history over which a difference can accumulate. The one-sign rows are a consistency check rather than evidence, since Corollary 3.10 makes the two couplings the same algorithm there; the two arms of that table were run on different seed streams, so the reported ratios are ratios of independent estimates of one quantity and their departure from 1 is sampling noise. The exact statement for that configuration is the matched-seed comparison of complete states reported in Section 3.5.

5 Conclusions

Averaging repeated simulations is a simple way to reduce sampling error for a signed gradient-particle solver. The mass signs help us choose how to couple those simulations. At a fixed pre-stage state, reflection maximizes an opposite-sign pair’s contribution to conditional variance, while synchronous increments minimize it; for a same-sign pair, reflection minimizes it. These pairwise statements hold simultaneously at every reconstruction point. The resulting switch leaves each simulation’s discrete law intact, so the averaged field retains the expectation and bias of the underlying solver. The coupling changes its sampling variance, and the reduction at a single diffusion stage has a closed form in terms of mass products and matched separations.

The choice of matching requires a separate comparison. Restricting pairs to sign classes allows reflection to be used throughout, and the pairwise argument alone cannot distinguish that construction from the spatially matched switch. In the principal Burgers experiment, within-sign matching captures most of the reduction relative to reflected rank pairing. The switched rule reduces the integrated variance further, to 0.739 [0.682, 0.801] of the within-sign value at 0.94 times its measured cost; its variance relative to reflected pairing is 0.127 [0.107, 0.151]. The profile and time comparisons support an advantage over within-sign pairing for three profiles at the two later observation times, and the flux comparison shows that the benefit also occurs with cubic transport. In held-out experiment B, the switched rule reaches the prescribed expected-error target with four trajectories, compared with six for within-sign pairing and thirty-six for reflected pairing. Experiment A provides a separate comparison with an unpaired ensemble and the RQMC adaptation at a different target and particle count.

The boundaries help identify where the construction is useful. When all gradient masses share one sign, switching gives exactly the same paths as reflected pairing, although pairing itself can still help relative to independent runs. No advantage over reflection is resolved in the tested separated-sign configuration, on the flat flanks of a profile, or for the bias-dominated peak-gradient functional. The total-error benefit also fades as discretization bias dominates, on a scale set by $B^* = V/b^2$. Conditional expectation at the final diffusion stage offers a complementary reduction: in the tested configurations it improves the switched estimator’s variance-times-cost further.

Three questions seem worth pursuing. Whether a terminal variance comparison holds under explicit resolution assumptions is open; Remark 3.11 rules out the unrestricted coordinatewise argument, while conditions such as $|m|h/\sigma \ll 1$ remain to be investigated. Whether the construction extends to solvers that sample a velocity label, where the coupling would have to act on the label draw as well as on the diffusion, is a design question rather than a corollary, and the relaxation gradient-particle schemes of Bertaglia, Pareschi and Caflisch [4] are the natural setting in which to ask it. Finally, the cost of the construction in a parallel implementation, where the two simulations must exchange matched masses at every stage, requires its own measurement.

Data and code availability

The manuscript source, figure-generation scripts, numerical evidence, and reproduction instructions are distributed with the accompanying repository at <https://github.com/stephen122204/sac-grw>. The evidence map records source files and statistical conventions for each reported result. Historical exploratory outputs are distinguished from the curated evidence used by this manuscript.

A Proofs

A.1 Proposition 3.3

Let $\Xi_k^A = (X_k^A, m_k^A, v_k^A)$ be the complete state entering stage k , with $k = 0, \dots, K-1$. Let $\mathcal{F}_k$ contain the joint history through the sort and velocity update of that stage, as in Definition 3.2. The deterministic part of Algorithm 1 acts on Ξ_k^A alone. We compare the conditional increment laws before constructing Ξ_{k+1}^A .

Replica A receives $\sigma Z^{(k)}$, with $Z^{(k)} \sim \mathcal{N}(0, I_N)$ independent of $\mathcal{F}_k$. Replica B receives

$$\Delta X_j^B = \sigma \varepsilon_{k,j} Z_{\pi_k(j)}^{(k)}.$$

Given $\mathcal{F}_k$, the matching and sign vector are fixed. A permutation of independent standard Gaussian coordinates preserves their joint law, and multiplication by fixed signs preserves both their distributions and independence. Therefore

$$\mathcal{L}(\Delta X^A \mid \mathcal{F}_k) = \mathcal{L}(\Delta X^B \mid \mathcal{F}_k) = \mathcal{N}(0, \sigma^2 I_N).$$

Each replica has the same conditional transition as the uncoupled solver. Induction over the stages proves equality of its complete trajectory law with the baseline law. The joint law of the two replicas changes because their increments are dependent. $\square$

Independence of the fresh vector and predictability of the rule justify conditioning before the draw. Bijectivity ensures independence within each replica, and unit-modulus multipliers preserve the variance. The particular rule (3.5) also needs nonzero masses. Retaining each particle's carried velocity in the state makes the deterministic update well defined.

A.2 Lemma 3.6

Both h_a and h_b vanish outside the intervals with endpoints a, c and b, c respectively, so the product is integrable. If a and b lie on the same side of c , say $c \leq a \leq b$, then $h_a = h_b = -1$

on $[c, a]$, $h_b = -1$ and $h_a = 0$ on $[a, b]$, and both vanish elsewhere, so $\int h_a h_b = |a - c|$, which equals $\frac{1}{2}(|a - c| + |b - c| - |a - b|)$ because $|b - c| = |a - c| + |b - a|$. The case $c \geq a \geq b$ is identical. If a and b lie on opposite sides of c the supports meet only at c and the integral vanishes, while $|a - b| = |a - c| + |b - c|$ makes the right-hand side vanish as well. $\square$

A.3 Proposition 3.7

Condition on the pre-final state and fix any reference point $c \in \mathbb{R}$. With h_a as in Lemma 3.6, the terminal reconstruction of simulation A satisfies

$$U^{(A)}(x) = u_- + \sum_i m_i^A h_{X_i^A}(x) + \left(\sum_i m_i^A \right) \mathbf{1}\{x \geq c\},$$

and likewise for B . The last term is deterministic, so it does not affect any variance or covariance; no condition on the total mass (2.4) is therefore needed. Setting $G_A = \sum_i m_i^A h_{X_i^A}$ and $G_B = \sum_j m_j^B h_{X_j^B}$, we may write $\bar{U}(x) = D(x) + \frac{1}{2}(G_A(x) + G_B(x))$, where $D(x)$ is deterministic under this conditioning. Consequently,

$$V_\bullet = \int_{\mathbb{R}} \text{Var} \bar{U} = \frac{1}{4} \left(\int \text{Var} G_A + \int \text{Var} G_B + 2 \int \text{Cov}(G_A, G_B) \right).$$

By Proposition 3.3 the first two integrals are the same for every admissible coupling, so only the covariance has to be evaluated.

Expanding and applying (3.6) to the coupled pair gives

$$\int_{\mathbb{R}} \mathbb{E}[G_A G_B] = \frac{1}{2} \sum_{i,j} m_i^A m_j^B \left(\mathbb{E}|X_i^A - c| + \mathbb{E}|X_j^B - c| - \mathbb{E}|X_i^A - X_j^B| \right).$$

For $\int \mathbb{E} G_A \mathbb{E} G_B$, introduce an independent copy $X_j'^B$ of X_j^B ; since $\mathbb{E} G_B = \mathbb{E} G_{B'}$ pointwise, the same expansion applies with X_j^B replaced by $X_j'^B$, and the first two groups of terms are unchanged because they involve only the marginals. Subtracting,

$$\int_{\mathbb{R}} \text{Cov}(G_A, G_B) = -\frac{1}{2} \sum_{i,j} m_i^A m_j^B \left[\mathbb{E}|X_i^A - X_j^B| - \mathbb{E}|X_i^A - X_j'^B| \right],$$

and $X_i^A - X_j'^B = (a_i - b_j) + \sigma\sqrt{2}\mathcal{N}(0, 1)$ in law, so the second expectation equals $g_{\sigma\sqrt{2}}(a_i - b_j)$.

For $i \neq j$ the increments applied to X_i^A and X_j^B are independent under REFLECT and SWITCH, so $X_i^A - X_j^B = (a_i - b_j) + \sigma\sqrt{2}\mathcal{N}(0, 1)$ in law as well and the bracket vanishes; only the diagonal survives. There, under REFLECT, $X_i^A - X_i^B = d_i + 2\sigma Z_i$ and the first expectation is $g_{2\sigma}(d_i)$. Under SWITCH the same holds when $m_i^A m_i^B > 0$, whereas when $m_i^A m_i^B < 0$ the two increments cancel exactly and the first expectation is $|d_i|$. Subtracting the two couplings,

$$\begin{aligned} \int \text{COV}_{\text{REFLECT}} - \int \text{COV}_{\text{SWITCH}} &= -\frac{1}{2} \sum_{m_i^A m_i^B < 0} m_i^A m_i^B [g_{2\sigma}(d_i) - |d_i|] \\ &= \frac{1}{2} \sum_{m_i^A m_i^B < 0} |m_i^A m_i^B| [g_{2\sigma}(d_i) - |d_i|], \end{aligned}$$

and multiplying by the factor $\frac{1}{2}$ from the decomposition of $V_\bullet$ gives (3.7). Non-negativity follows from Jensen's inequality,

$$g_s(d) = \mathbb{E}|d + sZ| \geq |\mathbb{E}(d + sZ)| = |d|. \quad \square$$

Remark A.1 (recovering the product form). When $\sum_i m_i^A = \sum_j m_j^B = 0$ the terms involving the reference point c cancel identically in the expansion of $\int \mathbb{E}[G_A G_B]$, and one obtains the product identity $\int_{\mathbb{R}} D_A D_B = -\frac{1}{2} \sum_{i,j} m_i^A m_j^B |X_i^A - X_j^B|$ quoted in Remark 3.8, with $D_\bullet$ the reconstruction measured from its left far-field value.

A.4 Numerical Checks of the Final-Stage Identity

The identity was checked by two routes that do not share the algebra of the proof. The first is a direct paired Monte Carlo over the final stage at six pre-final states drawn from the two-pulse problem, in which the conditional integrated variance of the pair mean is estimated by quadrature on a fine grid using the same draws for both couplings. All six states agreed with (3.7) within a nominal 95% interval built from 12 000 draws in 40 batches, and refining the quadrature window by a factor of 2.7 moved the estimate by 0.02%. The second evaluates the joint indicator probabilities directly: for a matched pair at a, b with $p = \Phi((x - a)/\sigma)$ and $q = \Phi((x - b)/\sigma)$, reflection gives $\Pr\{X_A \leq x, X_B \leq x\} = \max(p + q - 1, 0)$ and synchronization gives $\min(p, q)$, so that

$$V_{\text{REFLECT}} - V_{\text{SWITCH}} = \frac{1}{2} \int_{\mathbb{R}} \sum_{i: m_i^A m_i^B < 0} m_i^A m_i^B [\max(p_i + q_i - 1, 0) - \min(p_i, q_i)] dx$$

because like-sign pairs use reflection under both rules and contribute zero to the difference. Adaptive quadrature of this expression at a configuration with four particles of unequal mass magnitudes agrees with (3.7) to ten decimal places. The two expressions for the joint probabilities are the Fréchet–Hoeffding bounds of (3.2). This numerical check verifies their use in the integrated formula at that configuration; the argument in Section 3.1 establishes the pairwise result for all admissible bivariate couplings.

B A Two-Particle Monotonicity Counterexample

Take $u_- = 0$, $m = (1/2, -1/2)$, $h = 0.005$, $\nu = 10^{-10}$, and two steps. The initial positions are $(p, 0)$, with $p_L = -0.00275$ or $p_R = -0.00225$. Both initial carried velocities are $(1/2, 0)$. Without diffusion, the first transport puts the positive particle at -0.00025 in state L and $+0.00025$ in state R . Re-sorting therefore leaves the velocities $(1/2, 0)$ in L but assigns $(-1/2, 0)$ to the negative and positive particles, respectively, in R . After the second transport, the particle positions in identity order are

$$L : (0.00225, 0), \quad R : (0.00025, -0.0025).$$

At the probe $x_* = 0.00125$, the reconstructed fields are $-1/2$ and 0 .

Now restore the Gaussian displacements with $\sigma = 10^{-6}$. On the event that each of the four displacements of a trajectory has modulus at most $\delta = 10^{-4}$, neither its second transport ordering nor its terminal probe classification changes. The relevant ordering gap is $0.00025 > 2\delta$ and the smallest probe margin is $0.001 > 2\delta$. Hence the same field values hold on this event. For either initial state the exceptional probability is at most $\epsilon = 8\Phi(-100) < \exp(-5003)$. Since every field value is in $[-1/2, 1/2]$,

$$\mathbb{E}U_R(x_*, 2h) - \mathbb{E}U_L(x_*, 2h) \geq \frac{1}{2} - \frac{3}{2}\epsilon > 0.$$

The initial movement of a positive mass to the right thus increases the expected terminal field. This proves the claim in Remark 3.11 without inferring an expectation from a finite set of simulation seeds.

C Test Problems, Initialization and Reference Solutions

Initialization. For smooth profiles with equal far-field values, let

$$w_+(x) = \max(u'_0(x), 0), \quad w_-(x) = \max(-u'_0(x), 0).$$

Both have integral $\text{TV}(u_0)/2$. Their normalized cumulative distributions are

$$F_{\pm}(x) = \frac{2}{\text{TV}(u_0)} \int_{-\infty}^x w_{\pm}(y) dy.$$

For each sign class, we place $N/2$ particles at midpoint quantiles,

$$r_k = \frac{k - \frac{1}{2}}{N/2}, \quad X_{k,\pm}^0 = F_{\pm}^{-1}(r_k), \quad k = 1, \dots, N/2,$$

where $F_{\pm}^{-1}$ denotes the generalized inverse. The assigned masses are $\pm \text{TV}(u_0)/N$. This deterministic placement makes the initial state identical across replicas and comparison arms. For the main two-pulse experiments, the code approximates the cumulative integrals by the trapezoid rule on 400 001 points in $[-12, 12]$ and inverts them by linear interpolation. It assigns the same absolute mass to both sign classes.

The profile/time study uses a separate numerical initialization on 200 001 points in $[-6, 6]$. It differentiates the sampled initial profile, forms cumulative rectangle sums of its positive and negative parts, and inverts each cumulative sum by linear interpolation. If M_+ and M_- are those two numerical integrals, the assigned masses are

$$m_{k,+} = \frac{M_+}{N/2}, \quad m_{k,-} = -\frac{M_-}{N/2}.$$

The reconstruction constant is $u_0(-6)$. This construction has equal counts per sign class but does not enforce equality of M_+ and M_- . For the oscillatory profile, its total signed mass is approximately -2.34×10^{-7} . The profile/time comparisons are numerical comparisons of these discrete initial states, and do not invoke the zero-mass identity for the complete evolution.

Profiles. The smooth initial profiles are given in (4.1) and (4.2). The reported sign-region counts use a 40 001-point evaluation of u'_0 and a relative magnitude threshold of 10^{-6} . This gives 2, 3, and 13 regions for the bump, two-pulse, and oscillatory profiles. The counts refer to gradients above the numerical threshold, rather than to every zero of the analytic derivative on the real line.

The prescribed signed-particle configurations in Section 4.7 use N particles. The interleaved case alternates masses $\pm 1/N$ on $[-0.25, 0.25]$. The separated case places positive masses on $[-2.0, -1.4]$ and negative masses on $[1.4, 2.0]$. The merging case uses the intervals $[-0.9, -0.5]$ and $[0.5, 0.9]$. In the latter two cases there are $N/2$ particles of each sign, with magnitude $1/N$. The one-sign control instead assigns mass $-2/N$ to every particle on $[-1, 1]$.

References. For $f(u) = u^2/2$ and $f(u) = u^3/3$ the reference is a Fourier pseudospectral solution on $[-16, 16]$ with $M = 2048$ modes, $\frac{2}{3}$ -rule dealiasing and integrating-factor RK4 at $dt = 10^{-4}$. For Burgers, the Cole–Hopf transformation supplies an independent whole-line reference [8, 13]. With

$$F(y) = \int_0^y u_0(z) dz,$$

the transformed heat solution and the recovered field are

$$\begin{aligned} \phi(x, t) &= \frac{1}{\sqrt{4\pi\nu t}} \int_{\mathbb{R}} \exp\left(-\frac{(x-y)^2}{4\nu t} - \frac{F(y)}{2\nu}\right) dy, \\ u(x, t) &= -2\nu \frac{\partial_x \phi(x, t)}{\phi(x, t)}. \end{aligned}$$

The potential F is analytic for the Gaussian pulses. A stabilized evaluation agrees with the spectral solution to 5.57×10^{-14} in maximum norm at the reconstruction points. This is a numerical consistency check, not a certified quadrature error bound. For the cubic flux, halving dt and doubling M changes the solution by 1.6×10^{-13} . These reference calculations assess field error. They are not timed competitors and do not enter the fit of a control coefficient.

D Additional Estimator Constructions

This appendix specifies the supporting estimators used in Sections 4.3 and 4.5.1.

D.1 Conditional Evaluation and Known-Mean Controls

For conditional evaluation of the last diffusion stage, let (a_i, m_i) be one replica's pre-final configuration and let Φ denote the standard normal cumulative distribution function. Then

$$\mathbb{E}[\mathbf{1}\{a_i + \sigma Z_i \leq x\} \mid \mathcal{F}_{K-1}] = \Phi\left(\frac{x - a_i}{\sigma}\right).$$

The conditional field is therefore

$$u_- + \sum_i m_i \Phi\left(\frac{x - a_i}{\sigma}\right),$$

and we average this expression over the two replicas. This operation integrates out the final indicator noise, rather than replacing a random displacement by zero.

The heat control and frozen-velocity control use the Brownian displacement W_i accumulated by each particle identity. Their terminal particle positions are, respectively,

$$X_{i,H} = X_i^0 + W_i, \quad X_{i,F} = X_i^0 + v_i^0 T + W_i.$$

Here v_i^0 is the deterministic initial rank velocity, held fixed in the surrogate. The cumulative fields built from these positions have known means

$$\begin{aligned} \mu_H(x) &= u_- + \sum_i m_i \Phi\left(\frac{x - X_i^0}{\sqrt{2\nu T}}\right), \\ \mu_F(x) &= u_- + \sum_i m_i \Phi\left(\frac{x - X_i^0 - v_i^0 T}{\sqrt{2\nu T}}\right). \end{aligned}$$

To obtain these means, observe that the predictable sign and permutation transforms preserve the conditional Gaussian law of every fresh increment. Each identity's accumulated displacement consequently has distribution $\mathcal{N}(0, 2\nu T)$.

Writing $\bar{H}$ for the average heat-control field of the two replicas gives the control estimator

$$\bar{U}_c(x) = \bar{U}(x) - c(\bar{H}(x) - \mu_H(x)).$$

The frozen-velocity estimator is defined in the same way. We examine $c = 1$ and a fitted scalar coefficient. On a disjoint training sample, the latter is

$$c^* = \frac{\sum_\ell \widehat{\text{Cov}}(\bar{U}(x_\ell), \bar{H}(x_\ell))}{\sum_\ell \widehat{\text{Var}}(\bar{H}(x_\ell))}.$$

This minimizes the sample integrated variance over scalar coefficients. The common factor Δx cancels between numerator and denominator. We hold the fitted coefficient fixed during evaluation and report its training cost separately.

D.2 Randomized Quasi-Monte Carlo Implementation

One `scipy.stats.qmc.Sobol` engine with $d = 2$ and 30 bits is created per replicate and scrambled once. Successive time steps consume successive blocks of N points, as in his equation (45). SciPy's `scramble=True` is a left-matrix scramble combined with a digital random shift, not fully nested Owen scrambling [20, 27]. The first coordinate assigns a rank by $\lfloor Np_1 \rfloor$ and the second supplies the increment $\sigma\Phi^{-1}(p_2)$, with p_2 clipped to $[2^{-53}, 1 - 2^{-53}]$ and the clip count recorded. N must be a power of two. We do not claim this adaptation preserves the interacting discrete law. Its total error is measured independently, including its own bias. The rank-bijection assertion is checked outside the timed path.

E Uncertainty Methodology and its Calibration

Sample quantities and bootstrap intervals. For n independent pair fields, let $\langle \bar{U} \rangle$ denote their sample mean. The estimates corresponding to (2.11) are

$$\hat{V} = \frac{1}{n-1} \sum_{r=1}^n \|\bar{U}^{(r)} - \langle \bar{U} \rangle\|_{\Omega}^2,$$

$$\hat{b}^2 = \|\langle \bar{U} \rangle - u_{\text{ref}}\|_{\Omega}^2 - \frac{\hat{V}}{n}.$$

The subtraction removes the expected sampling contribution to the squared error of the estimated mean. The resulting estimate can be negative even though the population squared bias is nonnegative. We retain its sign in the planning analysis rather than replacing it by zero.

To form a bootstrap interval [9], we draw n realization indices with replacement and recompute the complete statistic, including both parts of any variance ratio. A paired comparison uses the same indices for its arms. Independent arms are resampled independently. The reported percentile interval consists of the 2.5th and 97.5th percentiles of the resampled statistic, with resample counts given in Section 4.1.

Intervals for held-out error. Let e_r be the integrated squared error of the complete estimator in block r , for $r = 1, \dots, 32$. We report the mean $\bar{e}$ and use the upper end of a Bonferroni-adjusted two-sided t interval,

$$U = \bar{e} + t_{31, 1-\alpha/(2J)} \frac{s_e}{\sqrt{32}}, \quad \alpha = 0.05,$$

where s_e is the sample standard deviation of the block errors and J is the number of arms in that experiment. Comparing U with the target assesses expected error under this interval approximation. It does not give the probability that an individual execution falls below the target.

Scalar variance-ratio calibration. The two interval procedures used for scalar observables are a whole-replicate percentile bootstrap and $F(n-1, n-1)$. The second assumes independent normal replicates, which we do not assert, so we calibrated both.

Per-replicate values of four observables were taken from the archived $N = 8192$ fields, separately for the reflected and sign-switched arms, and each sample was centered and scaled by its own standard deviation to give two empirical shapes. A synthetic numerator arm resamples the sign-switched shape, scaled to impose a known variance ratio r . The denominator arm resamples the reflected shape. Nominal-95% intervals are then formed by each procedure and coverage counted over $M = 1500$ synthetic datasets, with $n = 48$ per arm and 400 bootstrap resamples per interval. Table 9 reports the two-shape design together with the common-shape design of an earlier calibration, which used one standardized sample for both arms and therefore does not retain the observed difference between the two distributions.

Under the two-shape design, estimated F coverage is 0.949–0.982 and percentile-bootstrap coverage is 0.921–0.950, with simulation uncertainty as stated in the table. We therefore report F intervals for station and level-crossing variance ratios and bootstrap intervals for integrated-variance and mean-squared-error ratios. The calibration concerns scalar variance ratios. It does not establish coverage for integrated-field or MSE intervals. Those bootstrap intervals are approximate, particularly at these sample sizes. This describes performance under these empirical distributions at this configuration. It is not a guarantee for other problems, and a non-significant normality test records failure to reject, not normality.

Table 9 Coverage of nominal-95% variance-ratio intervals, $M = 1500$ synthetic datasets per cell, binomial standard error ≤ 0.007 . “Two-shape” resamples the sign-switched and reflected empirical shapes separately. “common-shape” reproduces the earlier design in which both arms came from one standardized sample.

observable	two-shape, $r = 0.15$		two-shape, $r = 0.75$		common-shape, $r = 0.75$	
	bootstrap	F	bootstrap	F	bootstrap	F
station $x = +0.1$	0.921	0.958	0.933	0.967	0.945	0.977
station $x = -2.0$	0.950	0.982	0.946	0.973	0.942	0.977
$\max \partial_x u $	0.941	0.965	0.931	0.960	0.947	0.954
level $x(u = 0.25)$	0.933	0.959	0.928	0.949	0.935	0.963

F Evidence Provenance

Table 10 maps every table and figure to the archived output that produced it. Paths are relative to the study repository’s `output/` directory unless stated otherwise. Each directory name carries a date suffix (`_2026_09_09` or `_2026_09_10`) which is omitted here for width. The file `analysis/numbers.json` contains each quantity together with the archive it was read from.

Table 10: Provenance of reported results.

item	what it answers	archive
Table 1, Fig. 2	principal field comparison	independent replay of <code>round22_corrections</code> ; <code>replayed_fields.npz</code> , <code>replay_audit.json</code>
cost figures, §4.2	what the coupling costs	<code>round22_corrections/costs.json</code> ; <code>round17_compare/compare.json</code> (pair versus single)
Table 2	against controls and a second flux	<code>round09_costs_control/fields.npz</code> , <code>.../rows.csv</code>
Table 3, Fig. 3 (left)	profiles and times	<code>round23_mechanism/mechanism.json</code>
Table 4, Fig. 3 (right)	matched-pair geometry	<code>round24_separation/separation.json</code> (<code>matched_state</code> rows; the <code>own_history</code> rows are the association-only variant, one of nine cells containing 1)
Table 5, Fig. 4 (top)	held-out work, experiment A	<code>round17_compare/compare.json</code>
Table 6, Fig. 4 (bottom)	held-out work, experiment B	<code>round23_mechanism/worktarget.json</code> ; interleaved timings from <code>round24_separation/timing_check.json</code>
station ratios, §4.6	where the benefit sits	<code>round18_observable/observable.json</code> , <code>.../f_intervals.json</code>
Table 7	nonlinear observables	replayed archive as above; empirical MSE by (4.4)
Fig. 5	bias-dominated regime	<code>round07_joint_cell/joint_cell.json</code>
Table 8	signed-structure map	<code>round21_applicability/applicability.json</code>
Prop. 3.7 check and share	is the identity right, and how much does it explain	<code>round25_finalstage/finalstage.json</code> ; aggregates from <code>round06_finalstep/finalstep.json</code>

Table 10 continued

item	what it answers	archive
Cor. 3.10, divergence fractions	when are the couplings identical	<code>round25_divergence/divergence.json</code> (200 seeds); <code>round22_corrections/corrections.json</code> (40 seeds)
Prop. 3.3 witness	does a replica match production	<code>round14_audit/audit.json</code> ; <code>single_vs_production</code> maximum absolute differences 0 in position, mass and field
Rem. 3.11	why there is no unconditional guarantee	<code>round07_counterexample/counterexample.json</code>
Table 9	are the intervals trustworthy	<code>round25_calibration/calibration.json</code> (supersedes the single-shape design in <code>round22_corrections/calibration.json</code>)

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